Non-Invasive BPM and Glucose Trend Estimator

Abstract

The **Non-Invasive BPM and Glucose Trend Estimator** is a smart health monitoring system designed to measure heart rate and estimate glucose trends without invasive procedures. The system uses an ESP32 microcontroller, MAX30105 pulse sensor, OLED display, and keypad interface to provide real-time monitoring and user interaction. A machine learning decision tree model is also integrated to predict possible health conditions based on user data and sensor readings.

The project aims to provide a low-cost, portable, and easy-to-use biomedical monitoring solution for educational and research purposes.

Introduction

Healthcare monitoring systems are becoming increasingly important in modern biomedical engineering. Traditional glucose monitoring methods require invasive blood sampling, which can be uncomfortable for users. This project focuses on developing a non-invasive prototype capable of monitoring heart rate and estimating glucose trends using sensor data and embedded processing.

The system continuously measures physiological signals and displays them on an OLED screen. Users can interact with the device through a keypad to access different monitoring modes and prediction functions.

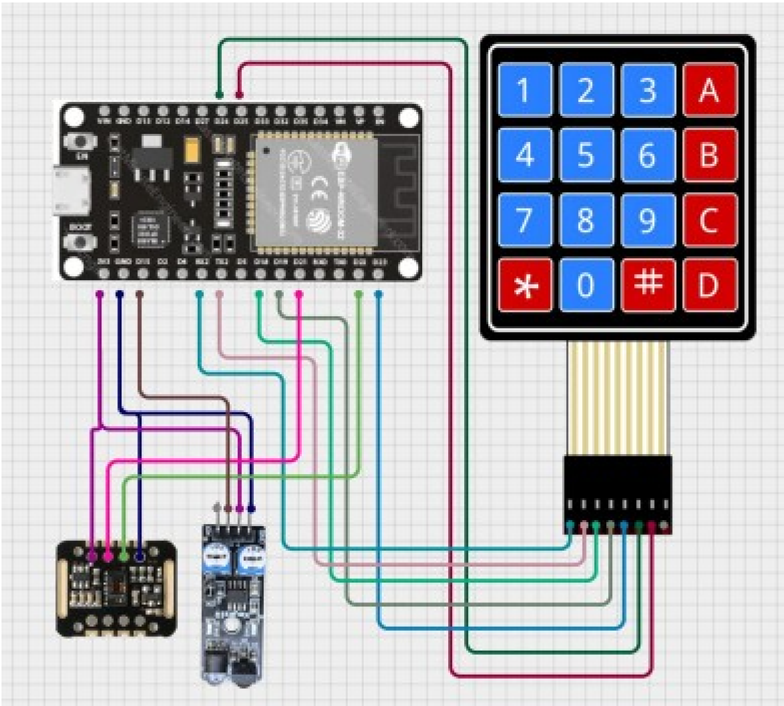
Objectives

The main objectives of this project are:

- To measure heart rate using the MAX30105 sensor.
- To estimate glucose trends using analog sensor readings.
- To display readings in real time on an OLED display.
- To implement a user-friendly keypad interface.
- To integrate a machine learning model for disease prediction.

Components Used

Component	Function
ESP32	Main controller
MAX30105 Sensor	Heart rate sensing
OLED Display	Display output
4x3 Keypad	User input
Analog Sensor	Glucose estimation



Working Principle

Heart Rate Monitoring

The MAX30105 sensor uses infrared and red light to detect blood flow variations in the finger. The ESP32 processes the sensor data and calculates heart rate values. To improve stability, moving average and exponential smoothing techniques are applied.

Glucose Trend Estimation

The analog sensor values are converted into estimated glucose levels within a specific range. Multiple samples are averaged to reduce fluctuations and improve consistency.

Display and User Interface

The OLED screen shows menus, sensor readings, and prediction results. The keypad allows users to:

- Select monitoring modes
- Enter personal data
- Navigate the system menus

Machine Learning Prediction

A decision tree machine learning model is used for basic disease prediction. The following inputs are used:

- Age
- Weight
- BMI
- Heart Rate
- Glucose Estimate

The model predicts possible conditions such as:

- Hypoglycemia
- Hyperglycemia
- Tachycardia
- Bradycardia
- Normal condition

The prediction result is displayed directly on the OLED screen.

Features

- Non-invasive monitoring
- Real-time heart rate measurement
- Glucose trend estimation
- OLED graphical display
- Machine learning integration
- Portable and low-cost design

Advantages

- Easy to operate
- Fast response
- Compact embedded system

- Useful for biomedical research
- Educational healthcare application

Limitations

- Glucose readings are estimated only
- Not suitable for clinical diagnosis
- Sensor readings may vary due to finger placement

Future Improvements

Future enhancements may include:

- Mobile app connectivity
- Cloud data storage
- Battery-powered operation
- Advanced AI prediction models
- Improved glucose estimation accuracy

Conclusion

The **Non-Invasive BPM and Glucose Trend Estimator** demonstrates the use of biomedical sensors, embedded systems, and machine learning in healthcare monitoring applications. The project successfully provides real-time heart rate monitoring, glucose trend estimation, and disease prediction in a portable and user-friendly device. It highlights the potential of smart non-invasive healthcare technologies for future medical systems.